CIE International A Level Physics

# Topic 21

# Alternating Currents

Revision Booklet

**This booklet covers:**

- Characteristics of Alternating Currents
- R.M.S. Values and Power
- Half-Wave Rectification
- Full-Wave (Bridge) Rectification
- Smoothing with a Capacitor

## Characteristics of Alternating Currents

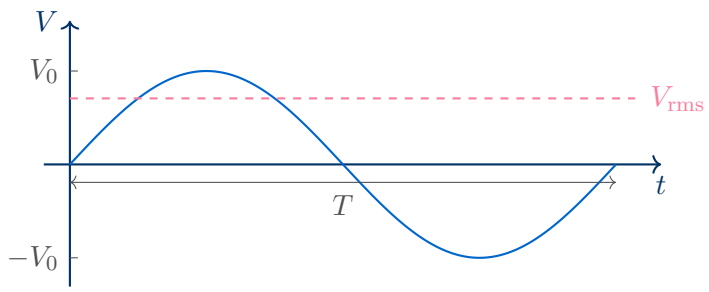
### Alternating Current

An **alternating current** (a.c.) reverses direction periodically. For a sinusoidal a.c.:

$$I = I_0 \sin \omega t \qquad V = V_0 \sin \omega t$$

- $I_0$, $V_0$: **peak** (amplitude) values.
- $\omega = 2\pi f = 2\pi/T$: angular frequency (rad s$^{-1}$).
- $T$: period (s); $f$: frequency (Hz).
- UK mains supply: $f = 50$ Hz, $T = 20$ ms, $V_{\text{rms}} = 230$ V.

**Sinusoidal a.c. voltage waveform**




## R.M.S. Values and Power

### Root-Mean-Square (R.M.S.) Value

The **r.m.s. value** of an alternating current is defined as the value of steady direct current that would dissipate the **same power** in a purely resistive load.

### R.M.S. Formulae

$$I_{\text{rms}} = \frac{I_0}{\sqrt{2}} \qquad V_{\text{rms}} = \frac{V_0}{\sqrt{2}}$$

These apply to **sinusoidal** waveforms only. For other waveforms the factor $1/\sqrt{2}$ changes.

### Mean Power in a Resistive Load

$$P_{\text{mean}} = \tfrac{1}{2}P_{\text{max}} = \tfrac{1}{2}I_0^2 R = I_{\text{rms}}^2 R = \frac{V_{\text{rms}}^2}{R}$$

The mean power is **half** the peak power for a sinusoidal waveform.

### Why Use R.M.S.?

- The mean value of a sinusoidal current is **zero** — useless for power calculations.
- R.M.S. values allow direct use of d.c. power formulae ($P = IV$, $P = I^2R$, $P = V^2/R$).
- Meters and ratings (e.g. 230 V mains) always quote r.m.s. values.

### Common Mistake

Do not use peak values in power calculations. Always convert to r.m.s. first: $V_{\text{rms}} = V_0/\sqrt{2}$. The peak mains voltage is $230\sqrt{2} \approx 325$ V, not 230 V.

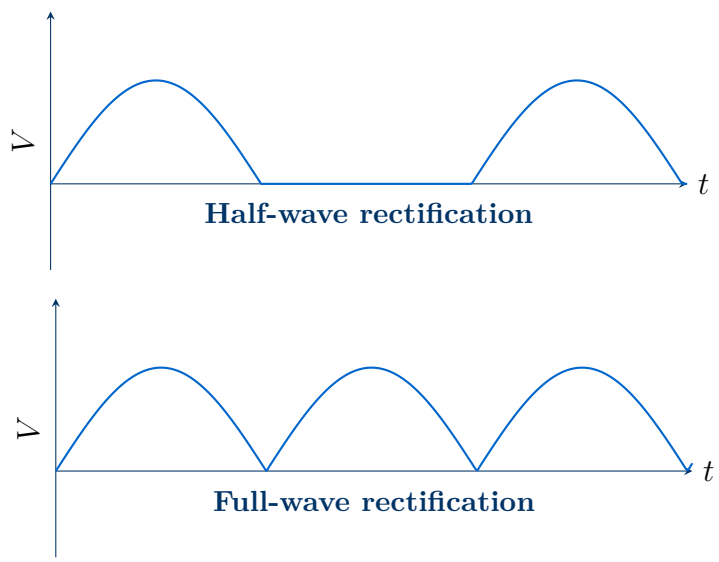
## Rectification

### Rectification

**Rectification** converts alternating current into direct current flowing in one direction only.

- **Half-wave rectification**: a single diode passes only the positive (or negative) half-cycles; output is a series of pulses with gaps.
- **Full-wave rectification**: a bridge rectifier (four diodes) inverts the negative half-cycles, producing a continuous pulsating d.c. output with no gaps.

**Comparison of rectified outputs**

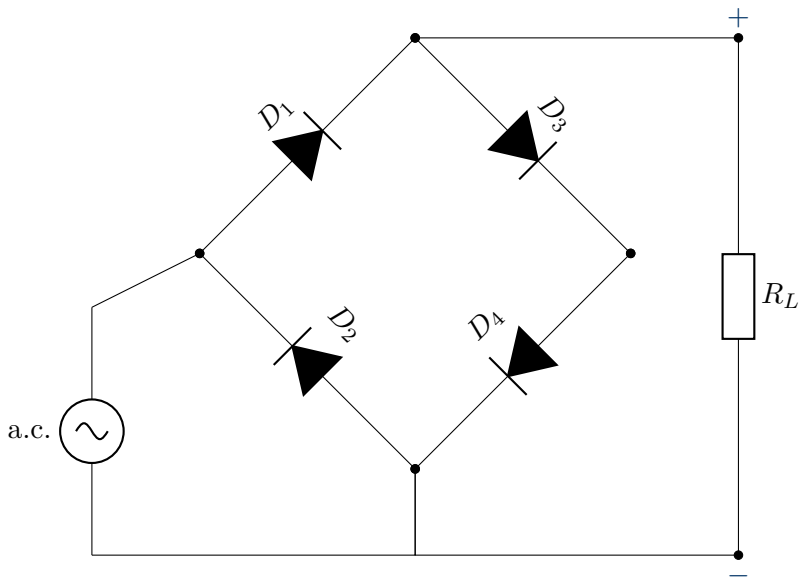


### Bridge Rectifier

**Bridge Rectifier**

A bridge rectifier uses **four diodes** arranged in a diamond so that:

- During the **positive half-cycle**: current flows through diodes D1 and D4, through the load in the positive direction.
- During the **negative half-cycle**: current flows through diodes D2 and D3, but still through the load in the **same** direction.
- Both half-cycles contribute to the output — no wasted half-cycles.



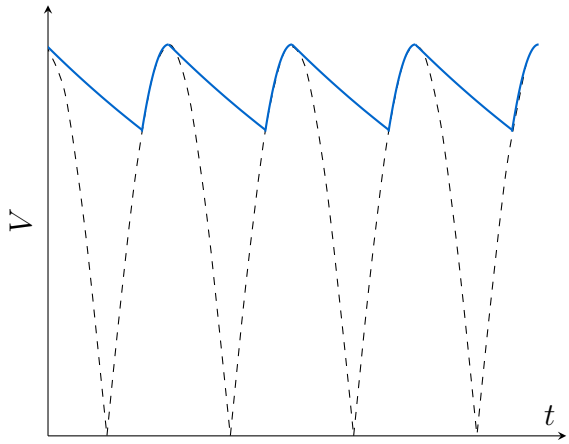

## Smoothing with a Capacitor

**Smoothing**

A capacitor connected in **parallel with the load** reduces the ripple on a rectified output:

- The capacitor **charges up** rapidly to the peak voltage during each pulse.
- It **discharges slowly** through the load $R_L$ between peaks, maintaining a more constant voltage.
- The output has a small residual **ripple voltage** rather than falling to zero between pulses.
- **Larger** $C$ or **larger** $R_L$: longer time constant $\tau = CR_L$, less ripple.
- **Smaller** $C$ or **smaller** $R_L$: faster discharge, larger ripple.

### Effect of smoothing capacitor on full-wave rectified output




**Common Mistake**

In exam questions, always link ripple size explicitly to the time constant $\tau = CR_L$ — just saying "bigger capacitor" without explanation will not earn full marks.

## Formula Summary Sheet

| Formula | Quantity | Units |
|---|---|---|
| $I = I_0 \sin \omega t$ | Sinusoidal alternating current | A |
| $V = V_0 \sin \omega t$ | Sinusoidal alternating voltage | V |
| $\omega = 2\pi f = 2\pi/T$ | Angular frequency | rad s$^{-1}$ |
| $I_{\text{rms}} = I_0/\sqrt{2}$ | R.M.S. current | A |
| $V_{\text{rms}} = V_0/\sqrt{2}$ | R.M.S. voltage | V |
| $P_{\text{mean}} = \frac{1}{2} I_0^2 R$ | Mean power (peak values) | W |
| $P_{\text{mean}} = I_{\text{rms}}^2 R = V_{\text{rms}}^2/R$ | Mean power (r.m.s. values) | W |

**UK mains:** $V_{\text{rms}} = 230$ V; $f = 50$ Hz; $V_0 = 230\sqrt{2} \approx 325$ V
**Note:** $1/\sqrt{2} \approx 0.707$; $P_{\text{mean}} = \frac{1}{2} P_{\text{max}}$ for sinusoidal waveform only.

## Worked Examples

### Example 1 — R.M.S. and Peak Values

**Question:** The mains supply has $V_{\text{rms}} = 230$ V at 50 Hz. Find (a) the peak voltage, (b) the angular frequency and (c) the mean power in a 1.2 kΩ resistor.

**Solution**

**(a)** $V_0 = V_{\text{rms}}\sqrt{2} = 230\sqrt{2} = \mathbf{325}$ V

**(b)** $\omega = 2\pi f = 2\pi \times 50 = \mathbf{314}$ rad s$^{-1}$

**(c)** $P = V_{\text{rms}}^2/R = 230^2/(1.2 \times 10^3) = 52900/1200 = \mathbf{44}$ W

### Example 2 — Peak Power and Mean Power

**Question:** An a.c. supply has peak voltage $V_0 = 12$ V and is connected to a 60 Ω resistor. Calculate (a) the peak power and (b) the mean power dissipated.

**Solution**

**(a)** $P_{\text{max}} = V_0^2/R = 144/60 = \mathbf{2.4}$ W

**(b)** $P_{\text{mean}} = \frac{1}{2}P_{\text{max}} = \mathbf{1.2}$ W

Or equivalently: $V_{\text{rms}} = 12/\sqrt{2}$; $P = V_{\text{rms}}^2/R = 72/60 = 1.2$ W

### Example 3 — Smoothing

**Question:** A full-wave rectifier feeds a 470 $\mu$F smoothing capacitor in parallel with a 2.2 kΩ load. Calculate the time constant and comment on the degree of smoothing for a 50 Hz supply.

**Solution**

$\tau = CR = 470 \times 10^{-6} \times 2200 = \mathbf{1.03}$ s

The period of the full-wave rectified signal is $T/2 = 1/(2 \times 50) = 10$ ms.

Since $\tau = 1.03$ s $\gg$ 10 ms, the capacitor discharges very little between peaks — the output is well smoothed with very small ripple.

## Practice Exam Questions

### Section A — Short Answer Questions

**Q1.** Explain what is meant by the r.m.s. value of an alternating current and state why it is more useful than the peak value.

*[3 marks]*

**Q2.** An a.c. supply has peak voltage 340 V. Calculate (a) the r.m.s. voltage and (b) the mean power delivered to a 680 Ω resistor.

*[3 marks]*

**Q3.** Distinguish between half-wave and full-wave rectification. State the number of diodes required for each.

*[3 marks]*

**Q4.** Explain how a capacitor connected in parallel with a load resistor smooths a rectified output. State the effect of increasing the capacitance.

*[3 marks]*

## Section B — Longer Structured Questions

**Q5.** The alternating voltage from a supply is given by $V = 170\sin(100\pi t)$, where $V$ is in volts and $t$ is in seconds.

(a) State the peak voltage and the frequency of the supply.

*[2 marks]*

(b) Calculate the r.m.s. voltage.

*[1 mark]*

(c) The supply is connected to a 500 Ω resistor. Calculate the mean power dissipated.

*[2 marks]*

(d) The supply is now passed through a bridge rectifier and a smoothing capacitor of 1000 $\mu$F is connected in parallel with the 500 Ω load. Calculate the time constant and comment on the effectiveness of smoothing.

*[3 marks]*

**Q6.** The graph below represents the output of a full-wave rectifier before smoothing. The peak voltage is 12 V and the supply frequency is 50 Hz.

(a) State the frequency of the rectified output.

*[1 mark]*

(b) On the same axes, sketch the output after connecting a large smoothing capacitor in parallel with the load. Indicate the approximate ripple voltage.

*[2 marks]*

(c) Explain what happens to the smoothing if the load resistance is reduced.

*[2 marks]*

## Mark Scheme and Answers

**Q1.** The r.m.s. value is the equivalent steady d.c. that dissipates the same power in a resistive load [1]; it is more useful because power formulae ($P = I^2R$, $P = V^2/R$) can be applied directly [1]; the mean of a sinusoidal current is zero, so it gives no information about power [1].

**Q2(a).** $V_{\text{rms}} = V_0/\sqrt{2} = 340/\sqrt{2} =$ **240** V [1]. **Q2(b).** $P = V_{\text{rms}}^2/R = 240^2/680 = 57600/680 =$ **85** W [2].

**Q3.** Half-wave: uses **1 diode**; only one half of each cycle is passed; output has gaps of zero voltage [1]. Full-wave: uses **4 diodes** (bridge rectifier); both half-cycles are used; output is always positive with no gaps [2].

**Q4.** The capacitor charges to the peak voltage during each pulse [1]; between pulses it discharges slowly through the load, maintaining a more constant output voltage [1]; increasing $C$ increases the time constant $\tau = CR$, so the capacitor discharges less between pulses and the ripple is smaller [1].

**Q5(a).** Peak voltage $V_0 = \mathbf{170}$ V [1]; $\omega = 100\pi$, so $f = \omega/2\pi = 100\pi/2\pi = \mathbf{50}$ Hz [1].

**Q5(b).** $V_{\text{rms}} = 170/\sqrt{2} = \mathbf{120}$ V [1].

**Q5(c).** $P = V_{\text{rms}}^2/R = (120)^2/500$ [1] $= 14400/500 = \mathbf{28.8}$ W [1].

**Q5(d).** $\tau = CR = 1000 \times 10^{-6} \times 500 = \mathbf{0.50}$ s [1]; period of full-wave output $= 1/(2 \times 50) = 10$ ms [1]; $\tau \gg T/2$ so capacitor barely discharges between peaks — very effective smoothing with tiny ripple [1].

**Q6(a).** The full-wave rectified output has frequency $2 \times 50 = \mathbf{100}$ Hz [1].

**Q6(b).** Sketch: smoothed output just below 12 V with small sawtooth ripple; ripple voltage is the small oscillation between the capacitor charge and discharge levels [2].

**Q6(c).** Reducing load resistance increases the discharge current [1]; the capacitor discharges faster between peaks ($\tau = CR$ decreases), so the ripple voltage increases and smoothing is less effective [1].

## Revision Checklist

Use this checklist to track your understanding. Tick each box when you are confident:

| | Learning Objective | Confidence (1–3) |
|---|---|---|
| ☐ | Define period, frequency and peak value for an a.c. waveform | |
| ☐ | Use $x = x_0 \sin \omega t$ for sinusoidal current or voltage | |
| ☐ | Define r.m.s. value and explain why it is useful | |
| ☐ | Use $I_{\text{rms}} = I_0/\sqrt{2}$ and $V_{\text{rms}} = V_0/\sqrt{2}$ | |
| ☐ | Calculate mean power using r.m.s. values: $P = I_{\text{rms}}^2 R = V_{\text{rms}}^2/R$ | |
| ☐ | State that mean power is half peak power for sinusoidal a.c. | |
| ☐ | Distinguish half-wave and full-wave rectification graphically | |
| ☐ | Explain the action of a single diode for half-wave rectification | |
| ☐ | Explain the action of a bridge rectifier (four diodes) | |
| ☐ | Explain smoothing by a capacitor in terms of charge/discharge | |
| ☐ | Analyse the effect of $C$ and $R_L$ on the time constant and ripple | |

*Key: 1 = Need more work 2 = Getting there 3 = Confident*

**Good luck with your revision!**

R.M.S. values are one of physics's most elegant ideas: a way to make a continuously varying quantity equivalent to a steady one. Once you see that $P_{\text{mean}} = \frac{1}{2}P_{\text{max}}$ comes directly from $\langle \sin^2 \rangle = \frac{1}{2}$, the whole topic falls into place.